

The positive generator for reflection-symmetric small-step walks

Candidate partial result for arXiv:2012.08947

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Status. Candidate substantial partial result, likely valid, pending expert review. The original conjecture remains open outside the family treated here.

Source question

Hoang, Raschel and Tarrago construct a distinguished basis $\{h_n\}_{n \geq 1}$ of killed harmonic functions in the quarter plane. Their Conjecture 1 asks whether the first generator h_1 spans the unique positive harmonic ray [1, p. 9].

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sense, positivity of the harmonic function corresponds to minimality in terms of polynomial degree.

Conjecture 1. *The unique (up to multiplicative factors) positive harmonic function associated to the Laplacian operator (H1)–(H5) is h_1 in Theorem 1.*

- The polynomial growth of the unique positive harmonic function can be read off on the corner point of the curve $\mathcal{Q}$ at 1 (see Figure 3) or equivalently on the

Transcription. “The unique (up to multiplicative factors) positive harmonic function associated to the Laplacian operator (H1)–(H5) is h_1 in Theorem 1.”

Definitions and result

Put $\mathcal{Q} = \mathbb{N}^2 = \{1, 2, \dots\}^2$ and kill the walk on leaving $\mathcal{Q}$. Consider the square-symmetric small-step law

$$p_{\pm 1, 0} = p_{0, \pm 1} = \alpha, \quad p_{\pm 1, \pm 1} = \beta, \quad p_{0, 0} = \gamma, \quad 4\alpha + 4\beta + \gamma = 1, \quad (1)$$

where $\alpha > 0$ and $\beta, \gamma \geq 0$; all unlisted probabilities vanish. Thus the law is invariant under both coordinate reflections and coordinate interchange. The assumption $\alpha > 0$ ensures irreducibility.

For the source kernel and generating functions we use

$$K(x, y) = xy - \sum_{k, \ell} p_{k, \ell} x^{1-k} y^{1-\ell},$$
$$H(x, y) = \sum_{i, j \geq 1} h(i, j) x^{i-1} y^{j-1}.$$

Theorem 1. *For every law (1), the line spanned by the source generator h_1 contains the function*

$$h(i, j) = ij \quad (i, j \geq 1), \quad h(i, j) = 0 \quad (i \leq 0 \text{ or } j \leq 0).$$

Consequently h_1 spans the unique positive killed harmonic ray, so Conjecture 1 of [1] holds for the whole family (1). This family contains the simple walk, the king walk, their mixtures, and all lazy versions.

Proof Intuition

Square symmetry kills exactly the three error terms in the one-step expansion of $(i+X)(j+Y)$, so ij is harmonic. The less obvious task is to show that it is the *particular* harmonic function produced by the source's conformal generator. Here the generating function of ij does that job for us: its kernel equation isolates a rational function $\psi(x)$. On the kernel curve, that identity says $\operatorname{Re} \psi = 0$. The rational function has the right double pole at the $\pi/2$ corner and no critical point in the kernel domain, so it is precisely the source Riemann map up to its harmless positive scale. The source formula for H_1 then collapses to the generating function of ij .

Proof

Let (X, Y) have law (1). Reflection invariance gives

$$\mathbb{E}X = \mathbb{E}Y = 0, \quad \mathbb{E}(XY) = \beta - \beta - \beta + \beta = 0.$$

Since the jumps are small, a step from $\mathcal{Q}$ can hit an axis but cannot overshoot it. The zero extension of $h(i, j) = ij$ therefore agrees at every possible endpoint with $(i+X)(j+Y)$, even when $i = 1$ or $j = 1$. Hence

$$\mathbb{E}h(i+X, j+Y) = ij + i\mathbb{E}Y + j\mathbb{E}X + \mathbb{E}(XY) = ij.$$

Thus h is positive and killed harmonic, and

$$H(x, y) = \frac{1}{(1-x)^2(1-y)^2}, \quad H(x, 0) = \frac{1}{(1-x)^2}, \quad H(0, 0) = 1. \quad (2)$$

The boundary pieces of the kernel are

$$K(x, 0) = -\alpha x - \beta(x^2 + 1), \quad K(0, 0) = -\beta.$$

Substituting (2) in the source functional equation

$$KH = K(x, 0)H(x, 0) + K(0, y)H(0, y) - K(0, 0)H(0, 0)$$

gives

$$K(x, y)H(x, y) = -\frac{\psi(x) + \psi(y)}{2}, \quad \psi(x) = \frac{\beta(x^2 + 1) + 2(\alpha + \beta)x}{(1-x)^2}. \quad (3)$$

Lemma 1. *The function ψ in (3) is, up to a positive multiplicative constant allowed by the normalization, the conformal coordinate ψ_1 of [1] from the bounded kernel domain $\mathcal{S}_1^+$ to the right half-plane.*

Proof. For $x \in \partial\mathcal{S}_1^+ = \mathcal{S}_1$, the source construction supplies the kernel pair $(x, \bar{x})$. Away from the corner 1, setting $K(x, \bar{x}) = 0$ in (3) yields

$$\psi(x) + \psi(\bar{x}) = 0.$$

The coefficients of ψ are real, so $\operatorname{Re} \psi(x) = 0$ on $\mathcal{S}_1 \setminus \{1\}$. Moreover,

$$\psi'(x) = \frac{2(\alpha + 2\beta)(x + 1)}{(1-x)^3}. \quad (4)$$

The Jordan domain $\mathcal{S}_1^+$ lies in the unit disk. Irreducibility excludes -1 from its boundary (a unit-modulus kernel point other than 1 would make the walk periodic in the sense excluded by source hypothesis (H3)). Thus (4) has no zero on $\overline{\mathcal{S}_1^+}$.

Square symmetry also gives mixed covariance $\sigma_{12} = 0$, so the corner angle in [1, (1.16)] is $\theta = \pi/2$. The sole singularity of ψ on the boundary is

$$\psi(x) \sim \frac{2\alpha + 4\beta}{(1-x)^2} \quad (x \rightarrow 1),$$

exactly the order $\pi/\theta = 2$ required at that corner. Because ψ' never vanishes, the zero set of $\operatorname{Re} \psi$ cannot acquire an additional arc in $\mathcal{S}_1^+$: such an arc cannot close (by the maximum principle), cannot end in the interior, and cannot meet the smooth boundary without locally coinciding with it. At the only nonsmooth point, the double pole and the

right-angle corner supply exactly the two boundary arms. Thus $\operatorname{Re} \psi$ has constant sign in $\mathcal{S}_1^+$. It is positive because $\psi(0) = \beta > 0$ when $\beta > 0$.

The map to $\mathcal{H}^+$ is proper: approaching the smooth boundary sends ψ to the imaginary axis, while approaching 1 sends $|\psi|$ to infinity. Its degree is especially easy to count. The equation $\psi(x) = \beta$ reduces to $2(\alpha + 2\beta)x = 0$, so 0 is its only finite solution. Hence the proper map has degree one. When $\beta = 0$, removing the holding probability leaves the simple-walk kernel curve and ψ is a positive multiple of $x/(1-x)^2$; for every positive regular value its two roots are reciprocal, with exactly one in the unit disk. The degree is again one. Hence ψ is conformal. Uniqueness of the source Riemann map gives $\psi_1 = c\psi$ for some $c > 0$ (with c fixed by whichever of the source's equivalent normalizations is used). $\square$

For $n = 1$, the source polynomial is $P_1(X) = X$, and Theorem 1 of [1] therefore gives

$$H_1(x, y) = \frac{\psi_1(x) + \psi_1(y)}{K(x, y)}.$$

Using Lemma 1 and (3),

$$H_1(x, y) = -2cH(x, y) = \frac{-2c}{(1-x)^2(1-y)^2}.$$

Coefficient extraction yields $h_1(i, j) = -2cij$. (The sign only records the source Riemann-map normalization; the conjecture is stated up to nonzero multiplicative factors.) Finally, finite support supplies every moment needed by the standard uniqueness theorem for the positive killed harmonic function invoked immediately before Conjecture 1 in [1]; see also [2]. Hence the line spanned by h_1 is exactly the unique positive ray. This proves Theorem 1. $\square$

Verification, limits, and novelty

The moment computation, all three kernel boundary terms, the derivative (4), and the coefficient extraction were checked separately. The cases $\beta = 0$ and $\gamma > 0$ were audited explicitly: holding only rescales the nontrivial harmonic equation. The main expert-review point is the degree-one argument in Lemma 1; its boundary values, unique pole, corner order, and absence of critical points are all displayed above.

The result does not cover asymmetric small-step laws or the source's walks with larger negative jumps. A bounded run-index and web/arXiv search on 27 August 2026 found the source's separate simple-walk and king-walk examples, but no explicit statement for the full family (1). Novelty is therefore plausible but provisional, not certified by an exhaustive search.

References

- [1] V. H. Hoang, K. Raschel and P. Tarrago, *Constructing discrete harmonic functions in wedges*, arXiv:2012.08947v2, 2023.
- [2] A. Bouaziz, S. Mustapha and M. Sifi, *Discrete harmonic functions on an orthant in $\mathbb{Z}^d$* , Electron. Commun. Probab. **20** (2015), no. 52, 13 pp.